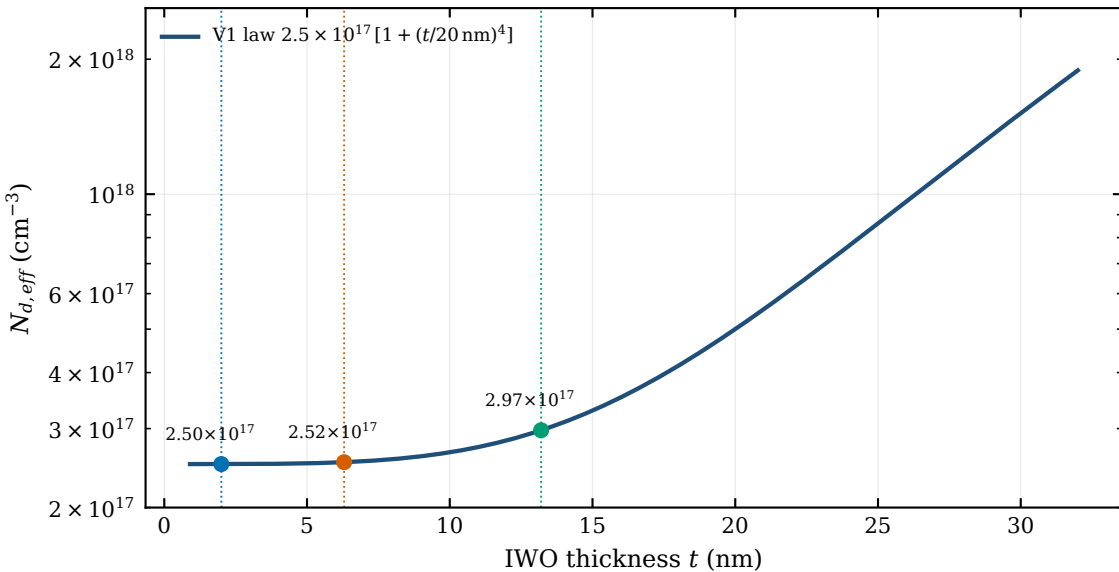


(d) Effective donor density $N_{d,eff}(t)$, model V1



Definition

$N_{d,eff}$ is the density of electrically active, fully ionized background donors, spread uniformly through the film (ATLAS DOPING, n-type). It lumps every net source of free electrons - W donors left after compensation, oxygen vacancies, hydrogen - into one number. It is NOT the W concentration: about 2 % W corresponds to roughly 10^{20} - 10^{21} cm^{-3} atoms, so almost all W donors must be compensated in these films.

Source of the curve

FITTED law (no direct measurement): nearly flat up to $\sim 10 \text{ nm}$ and steep beyond, to describe the thick, always-on 31.8 nm film. Its effect on the threshold is small for the thin films: $qN_{d,eff}t/C_{ox} = 0.009 / 0.028 / 0.070 \text{ V}$ at 2 / 6.3 / 13.2 nm. Qualitative support only: trap densities rising with thickness in a different IWO process [1].

References

[1] Kim et al., Appl. Phys. Lett. 125, 173507 (2024).